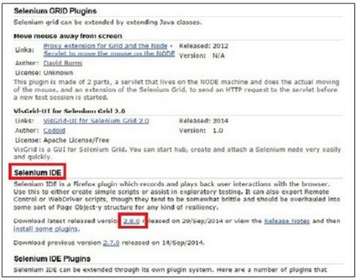

Figure 3: Selenium download page

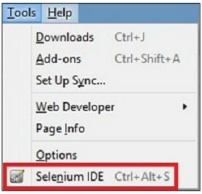

Figure 4: Selenium IDE Menu-Items

testing of software and Web applications. Various types of automation testing tools are available for the different tests required like regression, performance or functional testing.

One of the most commonly used automation tools is Selenium. It is one of the best options to test Web applications, and can be used for automating the testing of both the GUI as well as the features. It can also be used as a unit testing tool for JavaScript. It provides a record/playback facility to authorise the tests without knowing the test scripting language (Selenium IDE). It is implemented as a Firefox extension, which allows replaying, recording and editing test cases in Firefox.

Selenium allows testers and developers to develop test scripts to drive the browser. It prevents future regression of code; hence, it is popular. It can work on any browser that supports JavaScript, since it has been built using JavaScript.

Recorded test cases can be exported to various programming languages such as Java, C#, Python, Ruby, PHP and Perl.

Components of Selenium

Selenium is not just a single tool but a suite of products. It has four components—Selenium IDE, Selenium Grid, Web Driver and Selenium RC. Selenium RC and Web Driver have been merged to form a framework known as Selenium 2.

Selenium IDE: Selenium IDE is an integrated development environment for Selenium tests. It was developed by Shinya Kasatani. It allows you to use Selenium Core without having to copy it onto the server.

Selenium Core is a JavaScript module that allows Selenium to drive the browser. It can interact with Document Object Model (DOM) using JavaScript calls. Scripts are recorded in Selenese, a special scripting language for Selenium. It provides commands for performing actions in Web browsers and for retrieving data from multiple pages.

Selenium RC: This is a server written in Java, which accepts commands for the browser via HTTP. It makes it possible to write the tests for Web applications in any programming language. When using Selenium, a core user has to install the whole application under test alongwith the Web server on the local computer. So a server has been created by an engineer named Paul Hammant, which acts as a proxy to trick the browser into considering the Selenium core and Web application that is being tested as coming from the same domain. This is known as Selenium RC (remote control) or Selenium 1.

Selenium Web Driver: This is the successor to Selenium RC. It is the first cross-platform framework for testing that controls the browser from the OS level. It was created by Simon Stewart in 2006. It accepts commands,

Commonly used Selenium commands	
Command	**Description**
Open	Opens a page for the specified URL
Click	Clicks the elements (target) on the page
clickAndWait	Performs the click operation and waits for a new page to load
verifyTitle	Verifies the target title and continues to execute even if it fails
assertTitle	Verifies the target title and stops execution if it fails
verifyTextPresent	Verifies whether the specified text is present on the page
verifyElementPresent	Verifies whether the specified element is present by using its HTML tag
verifyText	Verifies whether the specified text and its corresponding HTML tag is present on the page
waitForPageToLoad	Pauses execution until an expected new page gets loaded
waitForElementPresent	Pauses execution until an expected UI element, as defined by its HTML tag, is present on the page